

Title:

Simulation tool for orchestrating quantum programs in distributed quantum environments.

Mentor:

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Abstract:

In the current NISQ era, it becomes important to combine and efficiently utilize multiple quantum processors for a collective use of computing power. In order to do so, we need an optimal strategy for efficient task distribution and resource utilization. In this project, we will develop a network aware orchestration and simulation tool for distributed systems of networked quantum devices. This will involve building interactive user interfaces and integrating networking capabilities in the open-source software called iQuantum. It will mainly involve JAVA based programming of various components. Such a tool can be very useful for the scientific community to test and develop algorithms for distributed orchestration of quantum programs in a networked system, without the use of costly hardware.

Research Plan (6 Weeks):**Week1:**

Installation, setup and understanding of iQuantum source code. Survey and understanding of core principles of quantum network characterization useful for our use cases.

Week2:

Design of code level architecture for network layer integration and interactive UI interfaces, including setup for quantum nodes environment, quantum programs characterization, network layer setup and result simulator.

Week3:

Development of network layer backend and frontend.

Week4:

Development of node and program characterization frontend. Unit testing for network level changes.

Week5:

Development of result simulator frontend and full integration testing.

Week6:

Documenting sample use cases and a user manual.

References:

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